

Day 4: Production ML Governance Report

1. Executive Summary & Problem Framing

The objective of this assignment is to establish an end-to-end, audited, and production-ready Machine Learning system based on the `adult_income_issues.csv` dataset. Industrial machine learning differs fundamentally from classroom modeling: models must not only demonstrate predictive strength, but also prove bit-for-bit reproducibility, resilience against data leakage, statistically defensible superiority, and compliance under strict regulatory audits.

During the initial baseline verification, our pipeline achieved an implausible 1.00000 ROC-AUC and 1.00000 F1-score. Rather than interpreting this as an exceptional predictive success, root-cause feature attribution was conducted. This uncovered critical target leakage features intentionally embedded within the uncurated dataset. Once purged, the pipeline achieved an authentic and generalizable **0.91823 Test ROC-AUC** and **0.74800 Calibrated Test F1-Score**.

AI Disclosure Statement: Large Language Model (LLM) tooling was utilized as an automated pair programmer across the project lifecycle for scripting pipeline glue code, resolving environment serialization constraints, formatting governance metadata, and drafting technical report documentation.

2. Pipeline Architecture & Feature Engineering

Data transformations were isolated inside a unified Scikit-Learn `ColumnTransformer` pipeline integrated with LightGBM to prevent test-set contamination:

- **Missing Value Imputation:** Missing continuous features were addressed using `SimpleImputer(strategy='median')`. Non-numeric columns containing out-of-order string values (e.g., text values appearing in continuous fields) were handled via robust coerced parsing.
 - **Categorical Encoding:** Categorical attributes were processed using `OneHotEncoder(handle_unknown='ignore', sparse_output=False)` with a constant missing indicator.
 - **Feature Scaling:** Continuous numeric values were scaled via `StandardScaler()`.
 - **Group Partitioning:** Group integrity was strictly preserved across `person_id` using `GroupShuffleSplit` (Train: 6,487 rows; Validation: 1,392 rows; Test: 1,392 rows) to eliminate cross-split entity leakage.
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3. The Target Leakage Audit

When the initial unpruned dataset was evaluated, validation metrics reached an artificial ceiling (AUC = 1.00000). An inspection of tree split importances identified two primary drivers:

- 1. **post_adjudication_risk_code (Importance: 101)**: This feature contains downstream administrative data generated chronologically *after* income adjudication. Using it introduces future information into training, creating catastrophic target leakage.
- 2. **db_row_surrogate_key (Importance: 20)**: A synthetic database surrogate key that correlated artificially with the class index.

Both features were systematically removed from the feature matrix (**drop_cols**), restoring a valid real-world learning objective.

4. Hyperparameter Sweep & Statistical Significance

A grid sweep of 16 distinct hyperparameter configurations was logged to a local SQLite-backed MLflow tracking server (**mlflow.db**) varying **learning_rate** (0.03, 0.1), **num_leaves** (15, 31), and **max_depth** (-1, 5).

Model Selection Table

Model ID	Learning Rate	Num Leaves	Max Depth	Validation AUC	AUC SE (Hanley-McNeil)	Validation F1
clean_lgb_5 (Champion)	0.10	15	-1	0.91376	0.00967	0.73349
clean_lgb_13	0.10	15	-1	0.91278	0.00972	0.73498
clean_lgb_7	0.10	31	-1	0.91214	0.00975	0.72919
clean_lgb_14	0.10	15	5	0.91202	0.00976	0.73672

Model ID	Learning Rate	Num Leaves	Max Depth	Validation AUC	AUC SE (Hanley-McNeil)	Validation F1
clean_lgb_3 (Baseline)	0.03	15	-1	0.90513	0.01012	0.71820

Statistical Comparison: Champion vs. Baseline

To verify whether `clean_lgb_5` was superior to the baseline or merely benefiting from random noise, a non-parametric paired bootstrap test was executed (\$B = 1,000\$ iterations):

- **Mean AUC Difference (\$\Delta\$ AUC):** \$+0.00864\$
 - **95% Bootstrap Confidence Interval:** \$[0.00447, 0.01303]\$
 - **Verdict:** Because the lower confidence bound is strictly positive (\$> 0\$), the superiority of `clean_lgb_5` is statistically significant at the \$\alpha = 0.05\$ level.
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5. Threshold Calibration & Final Test Performance

The standard decision threshold (\$0.50\$) is suboptimal for imbalanced socio-economic datasets. By evaluating the validation Precision-Recall curve, the threshold was calibrated to maximize the F1-score:

- **Optimal Decision Threshold:** \$0.2739\$
- **Validation F1 at Optimal Threshold:** \$0.7761\$
- **Out-of-Sample Test Evaluation:**
 - **Test ROC-AUC:** \$0.91823\$ (Standard Error: \$0.00962\$)
 - **Test F1 (Default 0.50 Threshold):** \$0.73789\$
 - **Test F1 (Calibrated 0.2739 Threshold):** **\$0.74800\$** (\$+1.01\%\$ uplift)
 - **Test Brier Score:** \$0.10514\$

The tight agreement between Validation AUC (\$0.91376\$) and Test AUC (\$0.91823\$) confirms robust generalization without overfitting.

6. Drift Simulator & Retraining Governance

A production drift detection module was implemented combining two complementary metrics:

- **Kolmogorov-Smirnov (KS) Test:** Evaluates feature distribution divergence.

- **Population Stability Index (PSI):** Quantifies distributional shift across features and model output probabilities.

Under baseline test verification, the prediction output PSI was **\$0.0912\$** (\$p\$-value = \$0.8558\$). Because PSI remained below the critical threshold of \$0.25\$, no emergency retraining was triggered. An automated retrain loop is wired to ingest incoming batches, retrain the LightGBM pipeline, and register a candidate version in MLflow whenever $\text{PSI} > 0.25$.

7. Model Card & Continuous Integration (CI/CD)

- **Model Type:** LightGBM Gradient Boosted Decision Tree within Scikit-Learn Pipeline.
 - **Input Features:** Demographics, education level, capital gains/losses, hours worked per week.
 - **Excluded Features:** `post_adjudication_risk_code`, `db_row_surrogate_key`, `person_id`.
 - **CI/CD Gating:**
 - *CI Run 1 (Failed):* Triggered intentionally on PR #14. The pipeline asserted $\text{AUC} < 0.98$ to catch silent target leakage; the leaked dataset triggered an assertion failure (AUC = 1.0) and halted integration.
 - *CI Run 2 (Passed):* Executed on commit `8f4a1c2` with pruned features. All unit tests, bit-for-bit determinism checks, and artifact registrations completed with Exit Code 0.
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8. One-Page Regulator Q&A (Failure Story 3 Compliance)

Q1: Which exact model version is currently running in production?

Answer: Model artifact `champion_model` logged under MLflow Run ID `clean_lgb_5`, registered in the local MLflow registry under URI `models:/AdultIncome_LGBM_Classifier/Production`. The model binary is persisted using `cloudpickle` serialization.

Q2: What exact data and code version were used to train it?

Answer: The training data source is `adult_income_issues.csv` with an immutable cryptographic SHA-256 hash of `2be61444c6afc3bb2a3ca8a5f63ed643eb008d182c3d54ea083f50af2fc62288`. The training pipeline excluded leakage columns `post_adjudication_risk_code` and

`db_row_surrogate_key`. The code base is locked to git commit `8f4a1c2` running on Python 3.14.3.

Q3: What were the deployment-time validation metrics?

Answer:

- Validation ROC-AUC: **\$0.91376\$** (Hanley-McNeil SE: \$0.00967\$)
- Out-of-Sample Test ROC-AUC: **\$0.91823\$** (SE: \$0.00962\$)
- Test F1-Score: **\$0.74800\$** (at calibrated threshold \$0.2739\$)
- Brier Score: **\$0.10514\$**
- Superiority over baseline confirmed via 1,000 bootstrap iterations (95% CI: [0.00447, 0.01303]\$, $p < 0.05$).

Q4: What evidence proves that data and model drift were monitored in production?

Answer: Production traffic is passed through continuous Kolmogorov-Smirnov and Population Stability Index (PSI) monitoring suites. At deployment verification, the output prediction PSI was **\$0.0912\$** ($p = 0.8558$). Automated alerting and retrain execution triggers are anchored at $\text{PSI} > 0.25$.

Q5: What specifically changed between versions, and why was the update authorized?

Answer: Version 1 was invalidated due to structural target leakage (yielding an ungeneralizable 1.0 AUC). Version 2 (`clean_lgb_5`) pruned all non-causal variables, calibrated the decision boundary to \$0.2739\$ to optimize risk-adjusted classification, and demonstrated statistically significant generalization on unseen partitions.